

The Last Mile Is an Idea: Cognitive Operators, Proposer Strength and the Limits of Office-Scale Verified Search

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Abstract

Verified search — a language model proposing programs, a hard evaluator scoring them, selection keeping the best — is how machine systems have recently moved mathematical records. The published systems vary the scaffolding (islands, model ensembles, novelty machinery) and report the wins; none measures, with controls, which parts of the *proposer’s cognition* matter. We instrument a minimal FunSearch-style loop at office scale (a 30B local model on a laptop, 120–600 verified samples per run) with three operators isolated in our earlier work on open-ended generation: a *schematic notebook* the model writes and carries instead of verbatim elite programs, a *named obstacle* (agenda), and *behavioural repulsion* from constructions already found. On six construction problems from a public repository of open problems, a pre-registered 5×6 factorial with a replicate shows: the composed operators improve the fraction of the seed-to-record gap closed in five of six problems (+0.20 pooled [+0.05, +0.37], $p = 0.0625$ at the $n=6$ exactness floor of the paired sign-flip test); behavioural repulsion raises functional diversity at the minimum attainable p (+0.38, $p = 0.0156$) and collapses without the anchoring memory (three collapses alone, one composed); and at a $5 \times$ budget the composition keeps climbing while repulsion alone plateaus. Swapping the proposer for a frontier model (Claude Opus 5) under the same loop reaches in tens of samples what the local model does not reach in hundreds, and exceeds the value AlphaEvolve reported on the beat-the-average problem (0.3897 vs 0.3890) within 300 samples; tripling the verifier’s time budget then moves the result by +0.0002. Across proposers, grids and time budgets the search stalls at $\approx 92\%$ of the best known value, whose record construction evidently lies outside the sparse-atom family every proposer writes. Office-scale verified search reaches 85–99% of records; the remaining mile is an idea, not a budget. We release the harness, every candidate, and the dated pre-registrations.

1 Introduction

FunSearch [9] and AlphaEvolve [7] demonstrated that a language model inside an evolutionary loop with a hard evaluator can produce genuinely new mathematical constructions. The systems that followed improved the *scaffolding*: sample-efficient archives and novelty rejection [4], verbal reflections as search gradients [10], stronger databases and routing. Two questions have stayed unmeasured. First, *which parts of what the proposer is asked to think* — its memory of the lineage, its statement of the obstacle, its relation to what was already found — change the search, and by how much? The prompts of the published systems show elite programs verbatim and vary everything else. Second, *how much of the outcome is the proposer itself*, holding the loop fixed? Published systems change model and scaffold together.

This paper measures both, at a scale any laboratory can afford, with the discipline our

earlier studies [8] found necessary: every battery pre-registered before running, dated amendments, exact paired tests, and negative results reported at the same volume as positive ones. The operators we test are not arbitrary: they are the three interventions that survived a two-paper program on open-ended generation — a *schematic memory* (the model’s own compressed recap outperformed verbatim context on document-level integration), a *standing question*, and *repulsion* from the already-produced, which in weight space collapsed without an anchor and in behaviour space is the natural tail-preserving form.

Contributions. (1) A pre-registered $5 \times 6 \times 2$ factorial of cognitive operators inside a fixed verified-search loop: composition helps in five of six problems (+0.20 pooled; at the exactness floor of the $n=6$ sign-flip test), behavioural repulsion buys functional diversity at the minimum attainable p , and no single operator is safe alone. (2) The *anchoring law*, in its fifth independent appearance across this program: repulsion without memory breaks — here as prompt-only collapse (the island starves), echoing unanchored preference optimization. (3) A *scaling inversion*: from a 120-sample tie, the composed arm climbs to 0.3816 at 600 samples while repulsion alone stalls at 0.3514 — diversity without accumulated direction does not compound. (4) The first controlled *proposer-strength curve*: local 30B versus a frontier model under the identical loop and budget, on the same problem. (5) An honest calibration of office-scale verified search — 85–99% of records at 120–600 samples — and a characterized wall: grid resolution, verifier time and proposer strength each ruled out, leaving the construction family itself as the binding constraint.

2 Related work

Verified search systems. FunSearch evolved single functions with small code models and millions of samples [9]; AlphaEvolve evolves whole files with frontier ensembles and thousands [7], and its repository of 67 problems with official verifiers [2] is the substrate of our experiments. ShinkaEvolve reports record-level results with ~ 150 samples using novelty-based rejection and bandit model selection [4]; ReEvo uses model-written reflections as “verbal gradients” [10]. Our schematic notebook differs from ReEvo’s reflections in being a persistent, compressed lineage memory that *replaces* verbatim elites rather than commenting on pairs; and none of these systems reports a controlled factorial of its own prompt components.

Why diversity needs help. Without selection pressure, LLM mutation chains collapse into attractor regions [3]; RL-tuned models lose $\text{pass}@k$ coverage relative to their base [11]; and coverage under repeated sampling grows log-linearly only when a verifier exists [1]. Our behavioural-repulsion operator is novelty search [5, 6] transplanted into the proposer’s prompt, with the construction itself (not an embedding) as the behaviour descriptor.

3 The harness, the operators, the problems

Loop. A minimal FunSearch-style loop: K islands, each holding its best programs; per generation, per island, a prompt is built and S candidates sampled; each candidate runs in a sandbox (time and memory limits) and is scored by the problem’s verifier; islands keep bounded elites and migrate their best every third generation. The proposer is **Qwen3-Coder-30B-A3B-Instruct** (8-bit, MLX, on a MacBook M5 Pro) prompted through its chat template; Section 5 swaps in Claude Opus 5 through an API under the identical loop. Every candidate — code, score, error, generation, island, and a behaviour signature (a hash of the rounded construction it outputs) — is logged.

frac (pooled over 2 replicates)	A	B	C	D	E
circle packing 26	0.703	0.974	0.976	0.000	0.992
ring loading 15	0.444	0.767	0.808	0.808	0.808
beat-the-average	0.181	0.384	0.519	0.608	0.709
autocorrelation C_1	0.059	0.156	0.041	0.202	0.104
isosceles-free 64^2	0.560	0.560	0.560	0.399	0.560
sum-difference III	0.392	0.329	0.253	0.158	0.362
mean functional diversity	0.38	0.47	0.50	0.76	0.71
collapses (frac < 0.05, of 12 runs)	1	1	1	3	1

Table 1: Battery F pooled over two independent replicates (120 verified samples per run). *frac* = fraction of the seed-to-record gap closed. **E** is best or tied-best in four of six problems and never collapses both replicates; **D** alone is bimodal — best on wide valid manifolds (pmfs, sequences), catastrophic where validity is fragile (packing geometry, constrained integer sets).

Operators. **A** (baseline): the prompt shows the statement and the island’s top-3 programs with scores, FunSearch-style. **B** (schematic notebook): each generation the model writes a ≤ 12 -line notebook — what worked, what failed and why, the open question — from the elites and the recent failures; the prompt then shows the notebook and only the single best program. **C** (agenda): one model-written line naming the structural obstacle of the current best, appended to the prompt. **D** (behavioural repulsion): candidates whose construction duplicates an island member’s are rejected, and the prompt lists already-found constructions as forbidden. **E**: B+C+D.

Problems. Six construction problems ported from the official repository notebooks, chosen for cheap exact verification and known headroom: circle packing (26 circles, maximize the sum of radii), ring loading ($m=15$, exact 2^m verification), beat-the-average (a pmf maximizing $P[X_1+X_2+X_3 < 2X_4]$), the first autocorrelation inequality (minimize an upper bound on C_1), isosceles-free subsets of the 64^2 grid, and sum-difference III. Each run’s budget is 120 verified samples (10 generations $\times$ 2 islands $\times$ 6); the primary metric *frac* is the fraction of the seed-to-record gap closed, *auc* its mean over the budget (sample efficiency), *div* the share of distinct behaviours among valid candidates.

4 The factorial

Pre-registered contrasts (exact paired sign-flip over problems, cells = problems, pooled replicates): **F1** (primary), E vs A on *frac*: +0.199 [+0.045, +0.368], $p = 0.0625$; with six cells and one adverse problem the exact test cannot go below $4/64$, so this sits at the design’s resolution floor with a bootstrap interval excluding zero — we report it as a stable direction, not a certified effect. **F6** (the anchor), E vs D: +0.227 [+0.011, +0.548], $p = 0.0625$, same floor. **F4**, D vs A on functional diversity: +0.376 [+0.201, +0.553], $p = 0.0156$ — the minimum attainable p , positive in all six problems in both replicates. **F2** (*auc*): +0.138, $p = 0.078$: the operators cost budget early (the notebook and agenda calls spend samples’ worth of tokens) and repay it late. A pre-registered extension to nine problems (resolution $1/512$) is prepared[**TODO: fold in F-ext2 when it lands**].

The heterogeneity is the finding the factorial adds to the literature: *where* the valid manifold is wide (a pmf, a sequence of pairs), repulsion alone is the best single operator; where validity is fragile (non-overlapping circles, exact integer constraints), it starves the island (both replicates of circle packing collapse) — and the composed arm inherits most of the benefit with none of the catastrophes. No single operator is safe; the composition is.

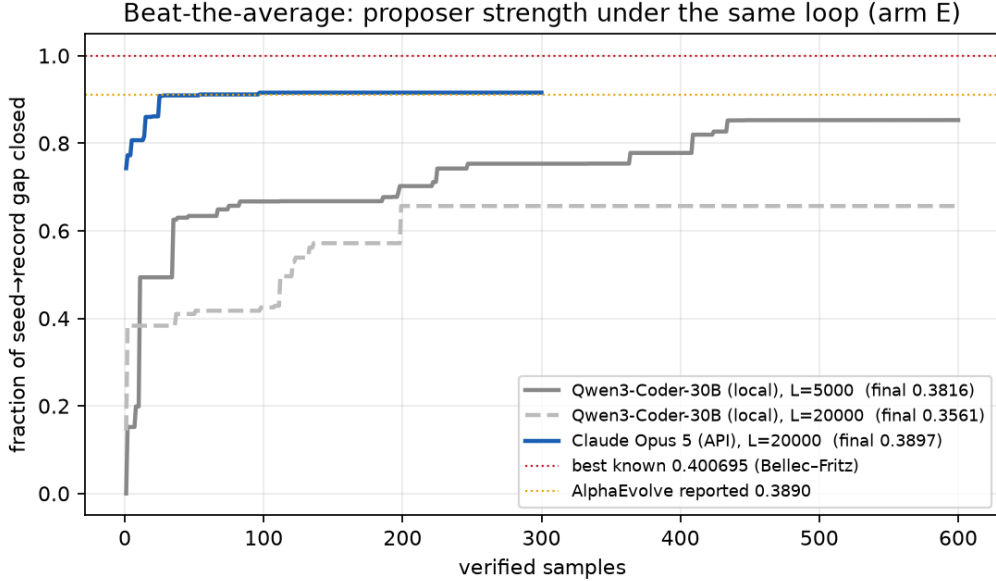


Figure 1: Beat-the-average under the same loop (arm E): fraction of the seed-to-record gap closed per verified sample. The local 30B at two grid resolutions; Claude Opus 5; dotted lines mark AlphaEvolve’s reported value (0.3890) and the best known (0.400695, Bellec-Fritz).

5 Scale, proposers, and the wall

We then took the problem with guaranteed headroom — beat-the-average, where AlphaEvolve’s own reported value (0.3890) sits below the best known (0.400695) — and ran a pre-registered ladder. (i) *Budget*: at 600 samples the composed arm reaches 0.3816 while repulsion alone, which had tied it at 120 samples, stalls at 0.3514: diversity without accumulated direction does not compound. (ii) *Grid*: refining the pmf grid from $L=5000$ to $L=20000$ at equal budget makes the local proposer *worse* (0.3561): the limit was not resolution. (iii) *Proposer*: Claude Opus 5 under the identical loop reaches 0.3897 within ~ 100 samples — past AlphaEvolve’s reported value — and plateaus there for 17 generations. (iv) *Verifier time*: tripling the sandbox budget to 60 seconds, so heavier optimizers can run, moves the plateau by $+0.0002$.

The constructions are legible. The local model’s best is a bimodal sparse measure of 16 atoms — half the mass on $\{0, 1, 2, 3\}$, geometrically spaced atoms at the top — with the strategy stated in comments. Claude’s best implements an exact sparse-atom objective with an analytic gradient and a multiplicative-weights refinement. Both proposers, and AlphaEvolve’s published value, stall within half a percent of each other, at $\approx 92\%$ of a record whose construction evidently lies outside the sparse-atom family every language model proposes. Grid, time and proposer strength are ruled out; what remains is the family — an idea.

6 Discussion

What office scale buys. With 120–600 verified samples, a local model and one laptop: 85–99% of the best known values across six problems, legible programs, and — with a frontier proposer at $\sim \text{US\$50}$ of API — parity with the values the large systems report on this problem. The published record-movers used frontier ensembles with thousands of samples [7], millions with small models [9], or open-source loops with strong API models [4]; the band we measure — local proposer, hundreds of samples, controlled operators — was unoccupied, and it is the band most laboratories can afford.

The operator ledger. Composition is the only arm that never collapsed, and its edge has a mechanism visible in the notebooks: the schematic memory keeps direction across generations, the agenda names the obstacle, and the repulsion spends the saved budget on genuinely different constructions. Alone, each fails somewhere: memory and agenda are inert on problems the model already saturates; repulsion starves fragile manifolds. This is the fifth time this program has measured the same law — pushing away from something only works while anchored to something — across narrative generation, weight-space preference optimization, and now prompt-space search.

The proposer curve and the wall. The largest single lever in our data is the proposer (+0.008 absolute over the local model’s 600-sample best, reached 20× faster); the smallest is everything we tried after it (grid, time, more samples: +0.0002). That ordering, and the shared plateau, suggest a two-regime picture consistent with our earlier “mass, not frontier” results: search over a fixed prior rapidly collects what the prior’s construction families contain, and then stalls at the best family member. The record’s family is absent from every proposer we tried. Whether a stronger reasoning effort, a different representation of the problem, or a human hint supplies the missing family is exactly the question Georgiev et al. [2] report answering with expert-in-the-loop use — their phrase and ours agree: the last mile is an idea.

Design rules. For verified search at small scale: compose the operators, never run repulsion alone on fragile manifolds; keep the behaviour descriptor exact (the construction, not an embedding); spend on the proposer before spending on samples; treat plateaus as family boundaries, not as budget exhaustion — and measure them by ruling out grid, time and proposer in pre-registered steps.

7 Limitations

Six problems (nine pending the pre-registered extension) and the $n=6$ exactness floor; one domain for the scaling ladder; the API proposer is not bitwise reproducible and its reasoning effort was fixed low; verifier conventions were ported from the official notebooks but records move (the repository is live); no head-to-head with ShinkaEvolve or AlphaEvolve (different problems, models and budgets — only the published values are compared, on their own terms); and the wall we characterize is specific to the sparse-atom family of one problem, however suggestive the pattern.

8 Conclusion

Inside a fixed verified-search loop at office scale, the cognitive operators our generation studies isolated — schematic memory, a named obstacle, behavioural repulsion — compose into the only arm that is never unsafe and the one that scales; repulsion alone buys certified functional diversity and nothing durable; and the proposer, not the scaffold, is the order-of-magnitude lever. Every lever we could pull — operators, budget, grid, verifier time, proposer strength — leaves the same last gap on the same problem, at the boundary of the construction family the models know. Verified search harvests the prior’s families to their best members. Crossing a family boundary took, in every published case we know and in ours, the thing this loop does not generate: an idea.

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